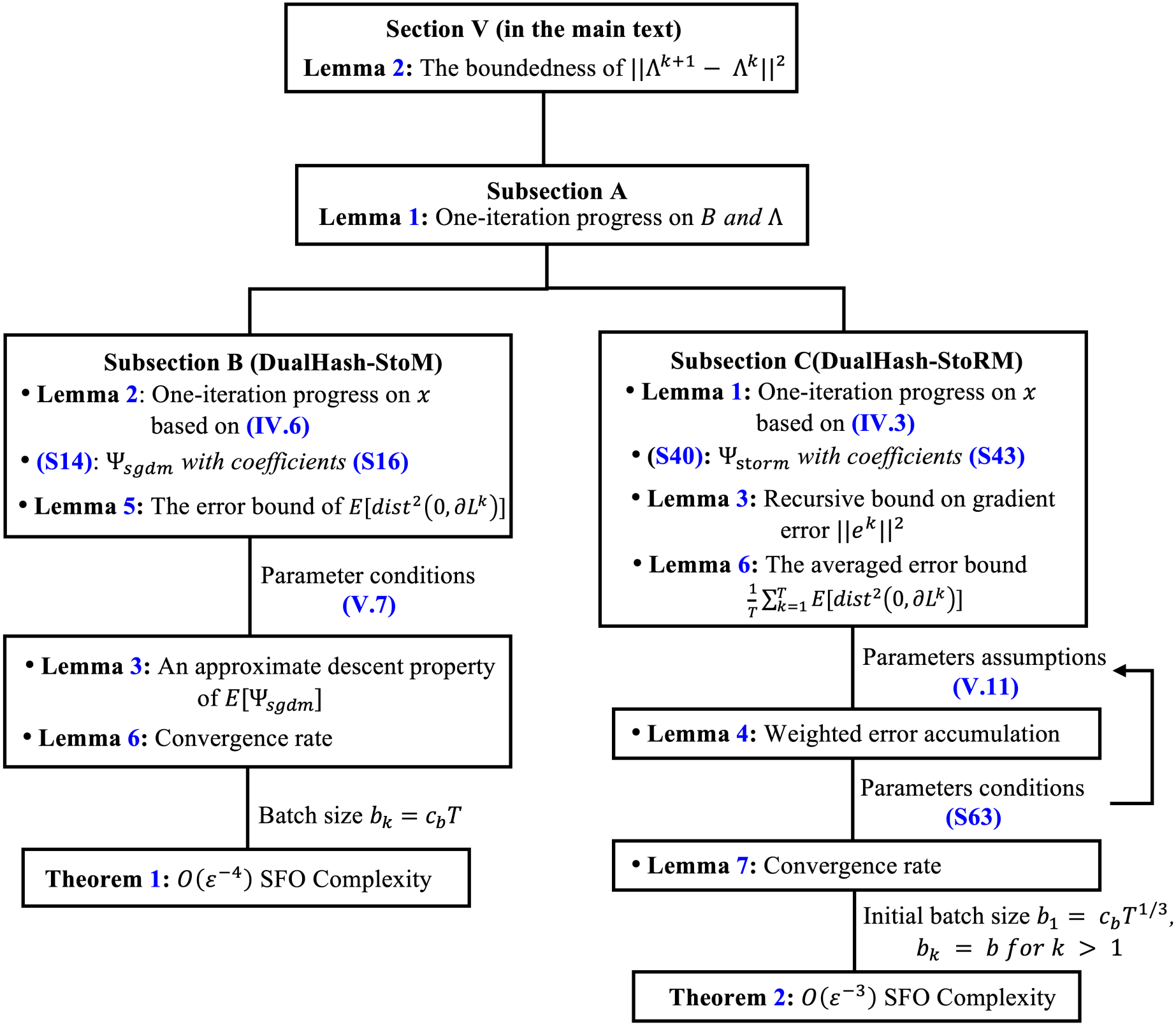
Section V (in the main text)
Lemma 2: The boundedness of $||\Lambda^{k+1} - \Lambda^k||^2$
Subsection A
Lemma 1: One-iteration progress on B and Λ
Subsection B (DualHash-StoM)
• Lemma 2: One-iteration progress on x based on (IV.6)
• (S14): Ψ_{sgdm} with coefficients (S16)
• Lemma 5: The error bound of $E[dist^2(0, \partial L^k)]$
Parameter conditions (V.7)
• Lemma 3: An approximate descent property of $E[\Psi_{sgdm}]$
• Lemma 6: Convergence rate
Batch size $b_k = c_b T$
Theorem 1: $O(\varepsilon^{-4})$ SFO Complexity
Subsection C(DualHash-StoRM)
• Lemma 1: One-iteration progress on x based on (IV.3)
• (S40): Ψ_{storm} with coefficients (S43)
• Lemma 3: Recursive bound on gradient error $||e^k||^2$
• Lemma 6: The averaged error bound $\frac{1}{T}\sum_{k=1}^{T} E[dist^2(0, \partial L^k)]$
Parameters assumptions (V.11)
• Lemma 4: Weighted error accumulation
Parameters conditions (S63)
• Lemma 7: Convergence rate
Initial batch size $b_1 = c_b T^{1/3}$, $b_k = b \; for \; k > 1$
Theorem 2: $O(\varepsilon^{-3})$ SFO Complexity